

Pagina 316

Esercizio 45.

$$L = \frac{F}{IB \sin \theta} = \frac{7.1 \times 10^{-5} \text{ N}}{0.66 \text{ A} \cdot 4.7 \times 10^{-5} \text{ T} \cdot \sin 58^\circ} \approx 2.7 \text{ m}$$

Esercizio 46.

$$\sin \theta = \frac{F}{ILB} = \frac{5.46 \text{ N}}{21.0 \text{ A} \cdot 0.655 \text{ m} \cdot 0.470 \text{ T}} \approx 0.845$$

$$\theta = \arcsin(\sin \theta) = \arcsin(0.845) \approx 57.6^\circ$$

Esercizio 47.

$$I_1 = \frac{F_1}{LB \sin \theta}$$

$$I_2 = \frac{I_1}{F_1} \cdot F_2 = \frac{2.7 \text{ A}}{30 \times 10^{-3} \text{ N}} \cdot 61 \times 10^{-3} \text{ N} \approx 5.5 \text{ A}$$
